

Simulation of Larval Feeding Routes in Social Wasps

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Abstract

This paper develops a multi-agent simulation to investigate larval feeding dynamics in primitively eusocial wasp colonies. The model represents rule-based interactions among foragers, feeders, and larvae on a two-dimensional grid. Adult wasps are assigned to two dynamic roles. Foragers operate near the nest periphery, collecting food at external foraging sites, whereas feeders move within the nest interior, receiving food from foragers and delivering it to larvae. We implement a set of hand-crafted movement strategies, including gradient-based greedy motion, uninformed and biased random walks, and local tour-planning heuristics inspired by traveling-salesperson routes. Colony-level behavior is summarized using larval hunger distributions, feeding efficiency, and feeding rates normalized by distance traveled and time steps. Initial results show that relatively simple local decision rules can produce uneven task participation and spatial feeding patterns similar to those reported for primitively eusocial wasps. The framework serves as a controllable baseline for adding further biological mechanisms and, in future work, learning-based decision policies.

Keywords— ecological simulation, wasp behavior, multi-agent systems, feeding dynamics

1. Introduction

Ecological systems often display complex group-level behavior that emerges from simple, local interactions among individuals, a theme extensively explored in systems modeling and ecology [1], [2]. Social wasps offer a clear example of decentralized coordination, in which foragers, feeders, and larvae exchange information and resources without rigid caste systems [3]. Many of these interactions occur inside the nest and are difficult to observe directly, motivating the use of computational simulation to study nest-level dynamics [4]–[6]. Agent-based ecological models reconstruct system behavior from rules specified at the individual level and have been applied to locomotion, task allocation, and resource distribution in social insects [7]–[11]. Designing such models requires balancing realism with interpretability. Highly detailed models can capture many biological nuances but are difficult to calibrate and analyze, whereas simpler models are interpretable but may omit important feedback mechanisms [14], [16]. Feeding dynamics, in particular, depend not only on local demand and food transfer between adults and larvae, but also on how adults move through the nest using strategies such as random walks, biased exploration, and multi-destination routing [17]–[22]. These behaviors interact with trophallaxis and encounter-driven task allocation among adults [24]–[27]. In this work, we construct a simulation of larval feeding routes in primitively eusocial wasps using a set of hand-crafted behavioral modes, including greedy selection, random walk, biased movement, and route-based heuristics. These movement rules provide controlled baselines for studying how individual-level decision policies generate colony-level patterns. A further objective is to compare simulation outputs with empirical feeding behavior documented for *Ropalidia marginata* by Sharma and Gadagkar [32]. Their study reports spatial biases in larval feeding, route optimization relative to random expectation, and inequality in task participation among adults. By benchmarking our simulation against these empirical metrics, we assess whether the modeled behaviors reproduce realistic nest-level patterns and identify where additional biological mechanisms may be required.

2. Problem Statement

Empirical studies of *Ropalidia marginata* have documented spatial biases in larval feeding, route-like adult movement, and unequal task participation among colony members. However, these dynamics are difficult to observe directly inside natural nests, and the biological mechanisms that produce them remain only partially understood. A central challenge is developing a computational model that can reconstruct these colony-level patterns well enough to support systematic experimentation.

Recent work by Sharma and Gadagkar suggests that larval feeding routes in primitively eusocial wasps may reflect decentralized solutions to multi-destination optimization problems. Building on this insight, our broader goal is to incorporate reinforcement learning into the decision-making of simulated wasps so that movement strategies can adapt to larval demand and environmental structure. Before such learning-based approaches can be meaningfully applied, there must first be a baseline simulation model that captures the core feeding dynamics of the real colony.

The problem addressed in this work is therefore twofold. First, we must construct a rule-based agent-based simulation that reproduces essential spatial and temporal signatures of feeding behavior observed in empirical studies. Second, we must evaluate whether simple hand-crafted movement heuristics are sufficient to approximate real feeding patterns or whether additional mechanisms, including learning-based decision policies, are necessary. Establishing this baseline is a prerequisite for integrating reinforcement learning into ecological simulations and for using such simulations to understand how real wasp colonies might organize efficient feeding behavior without centralized control.

3. Related Work

Ecological and systems theory has long examined how local interactions among components give rise to complex system-level behavior. Early modeling efforts demonstrated that feedback-driven processes play a central role in regulating ecological dynamics [1], [2], [4], [6], laying the groundwork for simulation-based analysis of biological systems that are difficult to observe directly.

Social insects provide a canonical domain for studying decentralized coordination. Classic work shows how colony-level behavior can emerge from simple interaction rules without strict caste hierarchies [3]. Subsequent studies have investigated self-organization, trophallaxis, encounter-based task allocation, and spatial interaction networks in social insects [4], [26]–[28].

Modeling movement and feeding dynamics requires capturing how agents search, traverse confined environments, and visit multiple targets. Prior research has described these behaviors using random walks, biased exploration, and multi-destination optimization frameworks such as shortest-path and Hamiltonian-style routes [17]–[22].

A persistent challenge in ecological simulation is balancing interpretability and predictive accuracy. Parametric models often demand careful calibration to reproduce realistic dynamics, leading to what has been described as an interpretability–complexity trade-off in ecological modeling [14], [16]. This trade-off is especially salient when models are used both to fit data and to generate mechanistic insight.

4. Methods

This section describes the wasp feeding simulator, including the agent-based environment, behavioral rules, feedback mechanisms, baseline pathfinding modes, and performance metrics used to characterize system-level behavior.

Simulator Overview:

We implement a custom Python-based multi-agent simulator that models feeding interactions inside a social wasp nest. The environment is represented as a two-dimensional grid containing fixed larval cells and walkable paths. The simulation proceeds in discrete time steps, during which adults move, exchange food, and update larval hunger. All agents act based only on local information, consistent with decentralized nest organization.

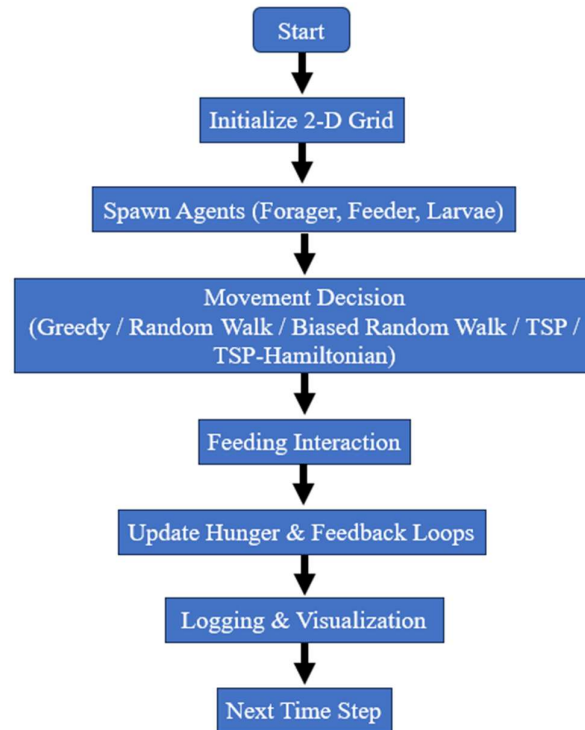


Figure 1. Overall simulation loop for the larval feeding model, showing grid initialization, agent spawning, movement decision, feeding interactions, hunger updates with feedback loops, and logging at each time step.

Agent Types and Behaviors:

The model includes two types of agents:

- **Adult wasps (Wasp agents)** with two dynamic roles
 - **Foragers:** operate near the nest periphery, collect food at external foraging sites, and can unload food to nearby feeders when local hunger cues are high.
 - **Feeders:** operate primarily inside the nest. They receive food from foragers or other feeders, move within the nest interior, and deliver food to nearby larvae. Feeders can also transfer food to other feeders when those feeders have high hunger.
- **Larvae:** remain at fixed locations, accumulate hunger over time, and generate demand signals that drive adult movement and feeding.

At each time step, movement is dictated by the selected pathfinding mode. Feeding occurs when an adult with stored food occupies a cell adjacent to a hungry larva or another hungry feeder.

Feedback Mechanisms and Role Switching:

The simulator incorporates several rule-based feedback interactions:

- **Larval demand → feeder allocation**
Larvae increase their hunger level over time. Hungry larvae contribute to a scalar field over the grid that attracts feeders more strongly toward regions with higher larval demand.
- **Forager → feeder food transfer**
When local hunger cues are high, foragers can unload food directly to nearby feeders, redistributing resources from the foraging ring into the nest interior.

- **Feeder → feeder food forwarding**

Feeders may transfer food to other feeders when those feeders are hungry, implementing a simple form of redistribution among adults.

- **Role switching based on hunger cues**

Feeders can switch to foragers when local hunger cues are high and the global fraction of foragers falls below a configurable threshold. Conversely, foragers can switch to feeders when perceived cues are low and the colony has an excess of foragers. A persistence counter limits how frequently individuals can change roles, preventing rapid oscillation.

These feedback mechanisms operate under all behavioral modes and provide the baseline dynamics of the system.

Baseline Pathfinding Algorithms:

We evaluate five handcrafted movement strategies that determine how adult wasps navigate the nest while delivering food to larvae.

- **Greedy (Gradient-Based) Movement**

Wasp movement is guided by a discretized gradient field constructed from larval hunger levels. At each step, the agent follows the sign of the local gradient components:

$$\Delta x_t = \text{sign} \left(\frac{\partial z}{\partial x} \right), \Delta y_t = \text{sign} \left(\frac{\partial z}{\partial y} \right).$$

Foragers without food move toward foraging sites, while food-carrying adults move toward hungry larvae. Feeders without food are also weakly attracted to nearby foragers.

- **Random Walk**

Agents select movement increments from the set $\{-1, 0, 1\} \times \{-1, 0, 1\}$ uniformly, producing unbiased exploration in the Moore neighborhood. Invalid moves (boundary violations or collisions) are resampled. The update rule is:

$$(x_{t+1}, y_{t+1}) = (x_t, y_t) + (\Delta x_t, \Delta y_t).$$

- **Biased Random Walk**

This variant is identical to the random walk except that steps returning directly to the previous location are rejected. This introduces mild directional persistence while maintaining unbiased neighborhood exploration.

- **TSP-Based Routing**

In the TSP mode, only feeder wasps perform local route planning. At each planning step, the feeder identifies all larvae within an interaction radius

$$R = r_{\text{smell}} + r_{\text{buffer}},$$

forming a local candidate set L_t . Based on its current food store food_t and the per-larva amount q , the number of larvae it can potentially serve is

$$m = \left\lfloor \frac{\text{food}_t}{q} \right\rfloor.$$

A subset $S_t \subseteq L_t$ is selected consisting of up to $\min(m, |L_t|)$ larvae that are closest to the feeder's current position X_t in Euclidean distance. An approximate open traveling-salesperson path visiting all nodes in S_t is then computed:

$$(v_1, v_2, \dots, v_k), v_i \in S_t.$$

The feeder connects to the tour via a shortest path in the local grid graph G :

$$P_{\text{start}} = \text{ShortestPath}_G(X_t, v_1).$$

The final planned route is the concatenation

$$P_t = P_{\text{start}} \parallel (v_1, v_2, \dots, v_k).$$

The wasp follows one node in P_t per simulation step. If collision-avoidance causes the wasp to drift more than one grid cell from the intended next node, the remaining route is discarded and replanning occurs in the next cycle.

- **Hamiltonian TSP Variant (Short Version with Equations)**

The Hamiltonian TSP mode uses the same planning steps as the open TSP strategy but constructs a closed tour. Over the same subset S_t , the algorithm computes an approximate Hamiltonian cycle

$$(v_1, v_2, \dots, v_k, v_1).$$

Connection to the cycle again uses

$$P_{\text{start}} = \text{ShortestPath}_G(X_t, v_1),$$

and the complete route is

$$P_t = P_{\text{start}} \parallel (v_1, v_2, \dots, v_k, v_1).$$

As in the open TSP mode, any significant deviation from the planned path triggers discarding of the remaining tour and recomputing a new local route.

5. Results

This section summarizes the system-level outcomes produced by the five baseline movement strategies: greedy gradient-based movement, random walk, biased random walk, TSP-based routing, and the Hamiltonian TSP variant. All simulations were run under identical initial conditions, and performance was evaluated using larval hunger distributions, feeding dynamics over time, and spatial variation in feeding success across distance bins. Together, these results illustrate how movement organization shapes colony-level feeding behavior.

Forager Recruitment and Local Hunger Cues:

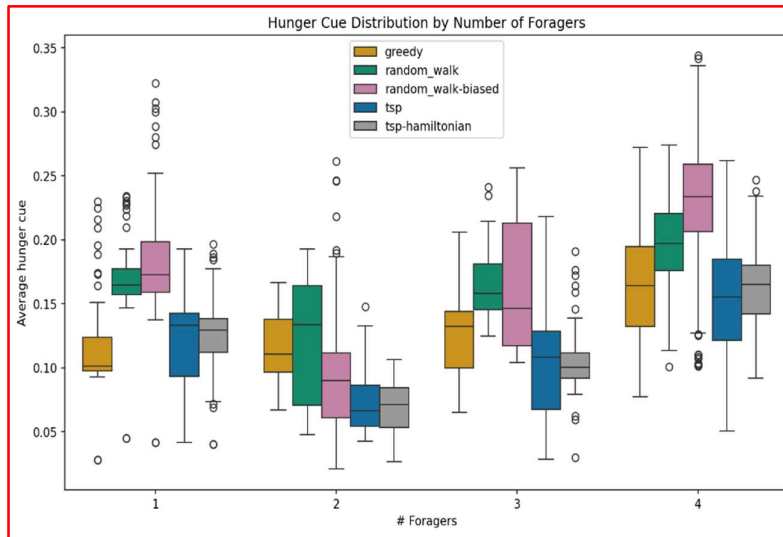


Figure 2. Forager Recruitment and Local Hunger Cues.

The relationship between local hunger cues and forager allocation is shown in Figure 2. Across movement modes, higher hunger cues are generally associated with larger numbers of foragers in the nest periphery. This trend reflects the role-switching rule in which adults transition into foraging when perceived hunger cues are high. The slightly elevated hunger level observed for the “one-forager” case is tied to the initialization phase before the simulation reaches steady-state, where early hunger accumulation temporarily exceeds the system’s capacity to redistribute food.

Overall, the results indicate that local larval demand influences the recruitment of foragers in a consistent manner across all pathfinding strategies.

Temporal Dynamics of Larval Hunger:

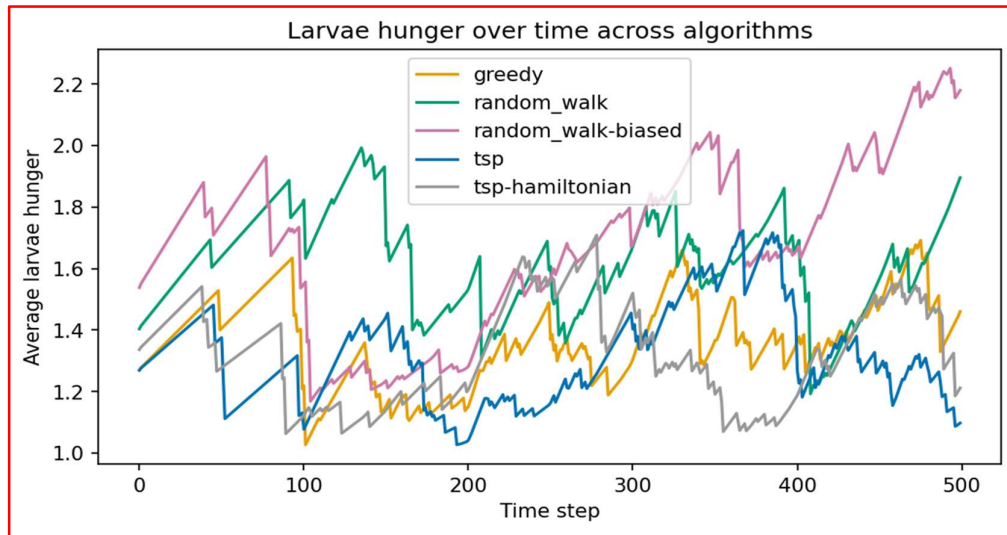


Figure 3. Temporal Dynamics of Larval Hunger.

Figure 3. shows the evolution of average larval hunger over the simulation period. All movement modes produce oscillatory, wave-like patterns reflecting repeated cycles of food delivery followed by gradual hunger accumulation. However, the amplitude and stability of these cycles vary substantially across modes. The TSP-based and greedy movement strategies maintain comparatively lower overall larval hunger throughout the simulation. Their more directed movement patterns enable feeders to reach and serve larvae more efficiently. In contrast, random-walk-based strategies exhibit larger fluctuations and higher peak hunger levels, reflecting their less structured spatial coverage. These patterns highlight how the predictability and structure of movement routes influence the temporal stability of feeding performance.

Spatial Patterns: Mean Larval Hunger Across Distance Bins:

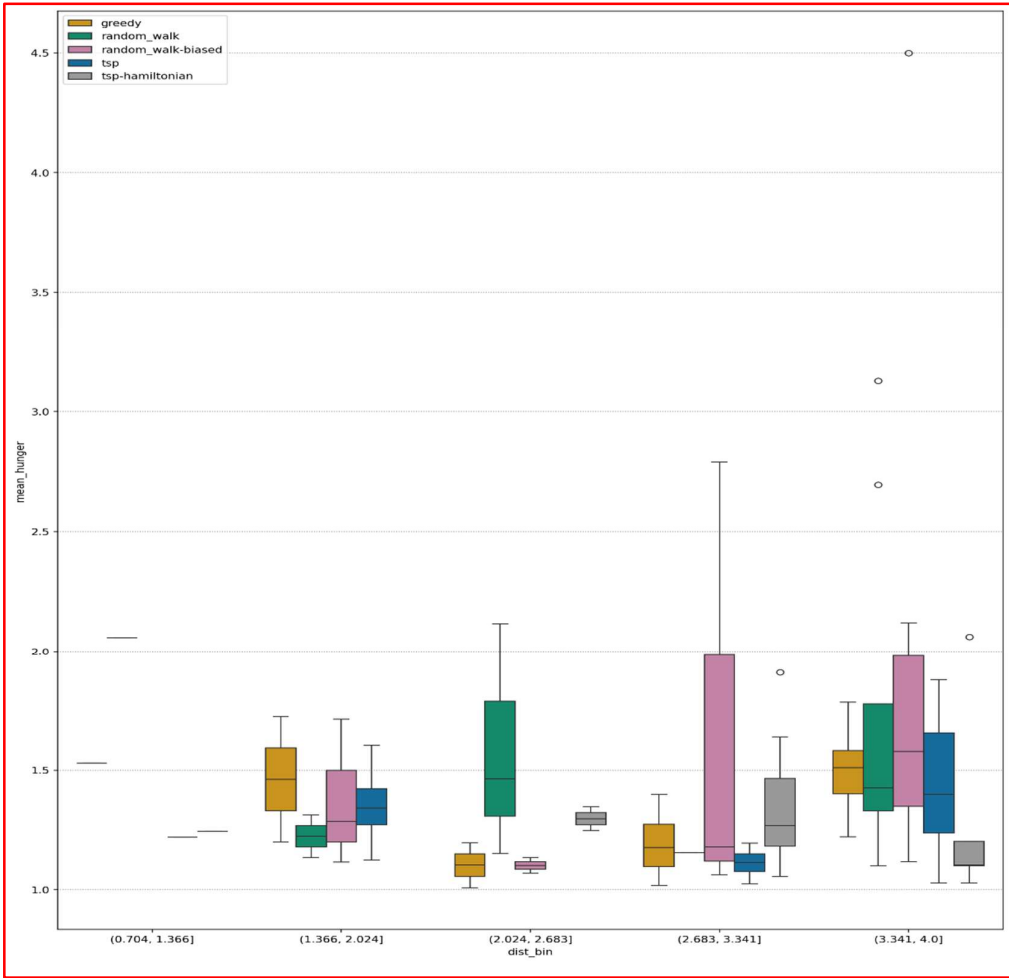


Figure 4. Spatial Patterns: Mean Larval Hunger Across Distance Bins.

Spatial variation in feeding success is illustrated in Figure 4. Larvae located in outer regions of the nest consistently receive less frequent feeding, resulting in higher average hunger levels across all strategies. This pattern is consistent with empirical observations reporting that peripheral larvae often experience reduced feeding consistency. A notable nuance is that larvae located near the geometric middle between the nest center and periphery tend to have the lowest hunger levels across movement modes. This reflects the natural flow of wasp traffic: central regions benefit from both incoming foragers and circulating feeders. Random-walk strategies also show the greatest variance across distance bins, indicating uneven spatial coverage and substantial variability in local feeding pressure. More structured strategies, such as TSP and TSP-Hamiltonian, reduce this variance by enforcing more complete coverage of larval positions.

Summary of Performance Trends:

Across all evaluated metrics, movement structure plays a significant role in determining feeding outcomes:

Directed and route-based strategies (TSP, Hamiltonian TSP): maintain lower larval hunger and provide more uniform spatial coverage.

Greedy movement: performs moderately well but exhibits higher variability, particularly at larger distances.

Random-walk strategies: produce the highest hunger levels and strongest spatial inequalities due to their undirected exploration patterns.

These results demonstrate that even in simple rule-driven systems, movement organization strongly modulates colony-level feeding performance.

6. Discussion

The comparative analysis of the five movement strategies demonstrates that the organization of adult movement plays a central role in determining colony-level feeding outcomes. Structured strategies such as the TSP and Hamiltonian-style routes consistently achieve lower larval hunger and more uniform spatial coverage than random-walk variants. Their ability to visit larval positions in a repeatable and predictable order reduces both redundant movement and localized starvation. Greedy movement, despite being responsive to larval demand, exhibits higher variability because it reacts only to local gradients and can over-focus on high-demand regions at the expense of global coverage.

Spatial inequality in feeding emerges as a robust pattern across all strategies. Larvae located in peripheral regions consistently experience higher hunger, mirroring empirical findings in *Ropalidia marginata* and suggesting that spatial geometry and flow constraints alone can bias feeding success. The lowest hunger levels occur among mid-distance larvae, where both foragers and feeders frequently pass through, indicating that movement flow—not only larval demand—shapes feeding opportunities.

The temporal dynamics further show that movement structure influences the stability of feeding cycles. TSP-based strategies generate smoother hunger oscillations, while random walks produce more pronounced fluctuations due to uneven coverage and increased likelihood of missing hungry larvae. These differences highlight that predictable movement patterns can stabilize resource distribution over time.

Overall, the findings support the idea that simple, locally executed rules can generate colony-level patterns resembling those observed in real wasp colonies. At the same time, the differences among strategies illustrate how movement heuristics mediate the balance between coverage, responsiveness, and spatial fairness. These insights help clarify which aspects of movement behavior are essential for reproducing realistic feeding dynamics and provide a foundation for developing more biologically grounded or learning-based decision rules in future work.

7. Limitations

While the simulation provides interpretable baselines for evaluating movement strategies in wasp colonies, several limitations should be acknowledged. First, the nest environment is modeled as a simplified two-dimensional grid with fixed larval positions and uniform movement costs. Real nests exhibit complex geometries, irregular cell spacing, and three-dimensional structures that may affect how adults navigate and allocate feeding effort. These structural features could amplify or suppress the spatial inequalities observed in this study.

Second, the behavioral rules used for foragers and feeders are intentionally stylized. Although they capture essential aspects of larval demand, food transfer, and role switching, the model does not include more nuanced biological mechanisms such as pheromone signaling, memory, individual variability, or cell-to-cell trophallaxis patterns documented in empirical studies. Similarly, the

gradient-based hunger field is a computational abstraction that may not fully capture how real larvae signal hunger or how adults perceive those cues.

Third, the pathfinding strategies—particularly the TSP and Hamiltonian variants—represent idealized heuristics that real wasps are unlikely to execute exactly. Their purpose is to benchmark coverage versus randomness rather than replicate biological movement. As such, care must be taken not to over-interpret direct comparisons between these artificial strategies and real wasp behavior.

Finally, simulation outcomes are sensitive to initial conditions and parameter choices, including hunger accumulation rates, transfer quantities, and agent-to-larva ratios. Although parameters were chosen to reflect typical scales reported in literature, full calibration against empirical data remains an open task. These limitations highlight opportunities for refining both the realism and the biological grounding of future models.

8. Future Work

Several directions can extend the present framework toward greater ecological realism and analytical depth. One promising direction is to incorporate more detailed nest geometries. Using actual nest maps from *Ropalidia marginata* or closely related species would allow more accurate evaluation of spatial constraints and peripheral underservice. Including heterogeneous movement costs or narrow passageways would also help capture realistic traffic patterns inside the nest.

A second direction involves expanding the behavioral rules to include additional biological mechanisms. Examples include pheromone-based signaling, probabilistic decision-making, individual variation among adults, and memory-driven route repetition. These mechanisms may help explain several empirical observations that are not reproduced by purely reactive or short-horizon movement strategies.

A third avenue for development is the exploration of learning-based decision policies. Deep reinforcement learning or multi-agent reinforcement learning can be used to compare hand-crafted heuristics with adaptive behaviors that optimize feeding efficiency, spatial fairness, or energetic cost. Such comparisons may reveal whether natural wasps behave more like optimized agents or rely on simpler evolutionarily shaped heuristics.

Another direction is to integrate empirical feeding logs, larval hunger measurements, or adult movement trajectories from observational studies. Incorporating real data would enable parameter calibration and statistical validation, and it would also strengthen the connection between the simulation and real colony behavior.

Finally, future work may investigate colony-level robustness by varying colony size, brood distribution, disturbance scenarios, or environmental conditions. These experiments could clarify how movement organization influences the stability and resilience of feeding networks when the colony is exposed to ecological stress.

9. Conclusion

This study demonstrates how baseline movement strategies influence larval feeding dynamics in primitively eusocial wasp colonies. Through an agent-based simulation incorporating foragers, feeders, larvae, and simple rule-driven interactions, we compare five pathfinding modes ranging from uninformed random walks to structured TSP-inspired routes. The results consistently show that

movement organization strongly shapes spatial coverage, temporal stability, and overall feeding efficiency.

Structured strategies such as TSP and Hamiltonian tours provide the most reliable feeding coverage and maintain lower larval hunger, while random walks generate higher variability and spatial inequality. The greedy gradient-following strategy, although responsive to larval demand, exhibits instability due to its purely local decision-making. These findings mirror several empirical patterns documented in *Ropalidia marginata*, supporting the notion that colony-level feeding outcomes can emerge from relatively simple individual rules.

By isolating the role of movement structure, the model establishes a foundation for integrating additional biological mechanisms or adaptive behaviors in future work. The framework serves as a stepping stone toward more realistic simulations that can bridge computational modeling with empirical behavioral ecology, ultimately improving our understanding of decentralized coordination in social insects.

Acknowledgements

The authors would like to express their sincere gratitude to Professor Nitika Sharma for her guidance, feedback, and continuous support throughout the development of this project. We also acknowledge the School of Information at the University of Arizona for providing the resources and academic environment that made this work possible. Finally, we thank our colleagues for their helpful discussions and encouragement during the simulation design and analysis process.

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